F 201-(33)

Devi Ahilya University
Institute of Engineering & Technology, Indore
Class Test: II

Sub Code: 3ETRG1/3EIRG1
Class: II Year
Duration: 70 Min.

Sub Name: Network Analysis
Branch: E & TC (A and B Section)/EI
Maximum Marks: 20

Note: All questions carry equal marks.

Q.1 Using Star-Delta transformation find the current through the galvanometer as given in fig 1. 05

Q.2 Capacitor is initially charged to voltage V_0 as shown in fig 2. At $t = 0$, switch K is closed. If $C_1 = C_2 = 1\ \mu F$, $V_0 = 10$ V and $R = 10\ \Omega$, find the (i) current i(t) (ii) voltage across R and (iii) charge across C_1 after $t = 10\ \mu S$. 05

Q.3 Describe Fourier transform and its symmetric properties. 05

Q.4 Find the Laplace transform of (i) $\sin \omega t$ and (ii) $e^{at} \cos \omega t$. 05

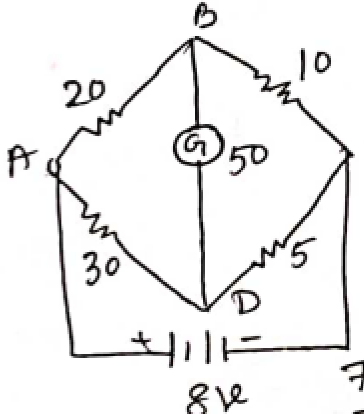

Fig-1

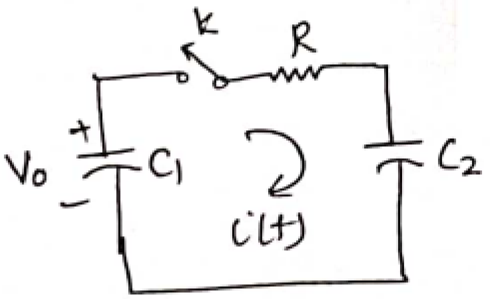

Fig-2